

GDSS-Maverick Hackathon

Title: AI-Driven Image-to-Item Master Data Tool (IMDB Auto-Fill from Product Images)

Problem

Retail teams currently fill the Item Master Database (IMDB) manually by transcribing data from product images or labels. This causes inconsistencies, duplication, and slow cataloging, especially for field teams, data scientists, and customer-service staff.

IMDB Definition and Target Columns

The IMDB is the central product-master table with the following 10 attributes: barcode, category type, segment type, manufacturer, brand, product name, weight and unit, packaging type, country of origin, and any useful promotional or marketing messages. The goal is to auto-fill all these fields from a product image.

Proposed Solution

This project builds an **image-to-IMDB tool** that:

1. accepts a product image (via mobile or web),
2. runs an image-analysis pipeline to extract the 10 IMDB attributes, and
3. outputs a **CSV or Excel file** that can be uploaded into a database or used as a product-master import table.

The system is **method-agnostic**: it may use:

- vision-language / multimodal large language models (VLMs),
- OCR-based text extraction with regex and pattern matching,
- multi-model pipelines combining object detection, logo recognition, and text extraction,
- or any other image-analysis method that robustly maps images to the 10 IMDB columns.

System Architecture (High-Level)

- Image ingestion and preprocessing (resize, enhance, focus on labels/barcodes).
- Attribute extraction (VLM, OCR, or multi-model) to fill the 10 IMDB fields.

- Validation and normalization (regex rules for barcodes, weights, country of origin, and branding).
- Export as **CSV / Excel file** (one row per product, one column per IMDB attribute) for later database upload.

Key Features (Aligned with Winning Criteria)

- **Accuracy & efficiency:** Auto-fill all 10 IMDB fields from images; flag low-confidence entries for human review.
- **Search-ready output:** Export structured data that supports filtering by brand, category, weight, and packaging once uploaded.
- **Centralized naming:** Standardize brand, category, segment, and packaging labels to reduce duplication.
- **Merge suggestions (bonus):** Compare new records with existing ones by barcode, brand, and weight to suggest potential duplicates.
- **User-friendly UI:** Simple upload → preview → edit → export workflow visible on mobile or web.

Process Plan

1. Select image-analysis method(s) and prepare sample product images (which will be provided).
2. Design prompts, OCR rules, and regex patterns for each IMDB column. Build validation logic for barcodes, weights, and country-of-origin.
3. Wrap into an end-to-end pipeline from image upload to Excel / CSV export.

Testing and Evaluation

- Compare human-filled IMDB records against model-filled ones on a test set.
- Measure accuracy per column (barcode, brand, weight, country of origin, etc.) and time saved vs. manual entry.
- Iterate on prompts and rules to improve weak columns.

Deliverables (Hackathon-Ready)

- A prototype UI and backend that:
 - accepts product images,
 - auto-fills the 10 IMDB columns,
 - exports a **CSV / Excel file** for database upload.
- A short demo showing: image upload → IMDB preview → export → illustration of how the file can be ingested into a product-master database.